Five metal particles (FMP) in UO_2: two-zone model

Model of V. Bellani (MSc thesis, Politecnico di Milano, 2026)
and changes EC November 2026, SCIANTIX 2.5

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4 November 2026

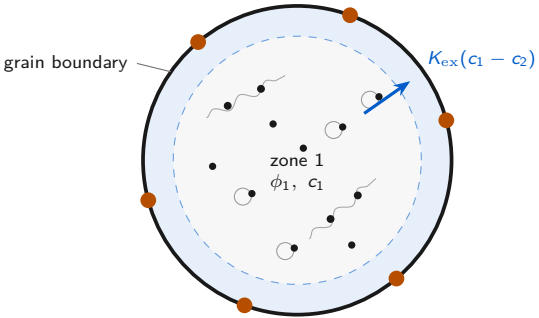
- black: model of V. Bellani, unchanged, with the equation number of the thesis [VB x.y]
- blue: new equation or assumption (EC November 2026)
- orange: modified with respect to V. Bellani (EC November 2026), the original form in grey

Code: `src/models/FiveMetalParticles.C`, `src/operations/SetPhaseDiagram.C`; cases and scripts: `validation/metallicinclusions/`; local branch `development/metallicinclusionsv2.5`.

Changes with respect to the model of V. Bellani

Item	V. Bellani	EC November 2026
Production	lumped yield $y = 0.578$ at/fiss [VB 3.3]	sum of the Mo, Tc, Ru, Rh, Pd systems, LWR yields, 0.5666 at/fiss
Matrix	one matrix, C_m [VB 3.4]	two zones: grain interior and grain-boundary zone ($\delta =$ 100 nm), exchange K_{ex}
Supersaturation	$S = C_m/C_{\mathrm{eq}}$ for both populations [VB 3.16]	$S_k = c_k/C_{\mathrm{eq}}$ of the zone of each population
Intragranular sites	$f_d\rho_d/\lambda + f_b N_b$ [VB 3.29]	same, times ϕ_1
Nucleation mass limit	both requests reduced by f_{mass} [VB 4.10]	each population limited by the metal of its zone
Implicit system	3×3 [VB 4.17]	4×4 with the exchange
Parameters	common k_0, B [VB 6.4, 6.5]	free k_0 and B for each population (new calibration)
Radius	from all the inventory [VB 3.49]	from the metallic part, $\phi_{\mathrm{met}} = 1 - x_{\mathrm{Mo}} f_{\mathrm{ox}}$
OpenCalphad input	— (all of "X produced" at the grain boundary)	grain-boundary inventory $x_X C^{\mathrm{GB}}$, Mo oxide fraction f_{ox}
Oxygen	not coupled	balance fuel / Mo oxides (Mo/MoO_2 buffer)
$f_{A,\mathrm{GB}}$	per fuel volume, not in the objective [VB 6.16, Sec. 6.3]	per volume of the grain-boundary zone, in the objective J_4
Calibration	2 parameters, $E_{\mathrm{cal}} = 0.614$ [VB 7.3]	4 parameters, $J_4 = 0.554$ ($J_3 = 0.598$)
Name, switch	5MP / MFP, `iCm`	FMP, `iFMP`

Two particle populations, two zones



zone 2: $\delta = 100$ nm on each side of the boundary (not to scale), ϕ_2, c_2

Intragranular particles N_{intra}, C^{intra}, R_{intra}

- sites: dislocations ($f_d \rho_d / \lambda$) and intragranular bubbles ($f_b N_b$)
- fed by the metal dissolved in zone 1, $c_1 = C_{m1}/\phi_1$

Grain-boundary particles N_{GB}, C^{GB}, R_{GB}

- sites: the boundary surface, $3/(2a\lambda^2)$
- fed by the metal dissolved in zone 2, $c_2 = C_{m2}/\phi_2$

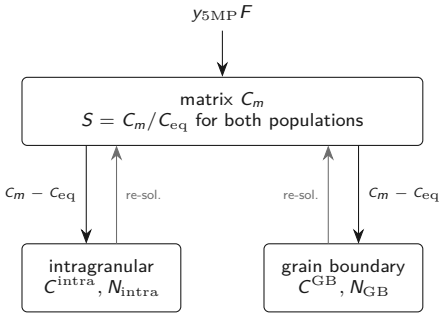
Processes

- production $\phi_k P$ in each zone; nucleation (CNT); diffusion-limited growth
- re-solution, back to the zone of the particles
- exchange between the zones, $K_{ex}(c_1 - c_2)$

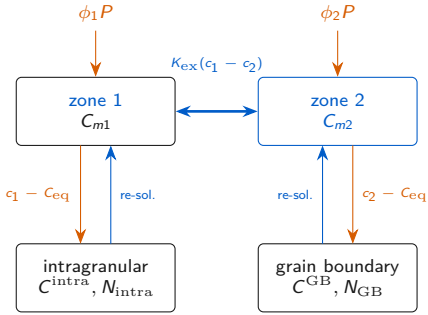
V. Bellani: one matrix C_m feeds both populations with the same supersaturation (thesis Fig. 3.1): with common parameters $R_{\mathrm{GB}} = R_{\mathrm{intra}}$ (thesis Tab. 7.2).

Metal balances: one matrix vs two zones

V. Bellani (thesis Fig. 3.1, eqs. 3.5–3.7)



Two zones (EC November 2026)



Total inventory conserved in both: production is the only net source [VB 3.8]. With the exchange the two populations see different concentrations, so the grain-boundary parameters can differ from the intragranular ones.

State variables

Symbol	Meaning	Unit	SCIANTIX variable
$C_{m1} + C_{m2}$	metal dissolved in the UO_2 matrix [VB Tab. 3.1, C_m]	at/m^3	`FMP in matrix`
C_{m2}	metal dissolved in the grain-boundary zone	at/m^3	`FMP in grain-boundary zone matrix`
C^{intra}	metal in intragranular particles	at/m^3	`FMP in intragranular particles`
C^{GB}	metal in grain-boundary particles	at/m^3	`FMP at grain boundary`
N_{intra}, N_{GB}	particle number densities (per fuel volume)	FMP/m^3	`Intra-/Intergranular FMP concentration`
R_{intra}, R_{GB}	mean radii, output (VB: atoms per particle)	m	`Intra-/Intergranular FMP radius`
ϕ_2	volume fraction of the grain-boundary zone	–	`FMP grain-boundary zone volume fraction`
O_{ox}	oxygen in the Mo oxides of the particles	at/m^3	`FMP oxygen in Mo oxides`
f_{ox}	fraction of the grain-boundary Mo in oxide phases (OpenCalphad)	–	`Mo in oxide fraction`

Inputs from SCIANTIX: temperature T, fission rate F, burnup b, grain radius a, intragranular bubble concentration N_b (one-way coupling, [VB Sec. 3.2]). The five metals are one lumped species [VB Sec. 3.2].

Production and inventory

V. Bellani: lumped yield from the PRODHEL composition (Angleraud), $x_{5\mathrm{MP}} = 0.289$ [VB 3.2]

$$\dot{C}_{5\mathrm{MP}} = y_{5\mathrm{MP}} F, \qquad y_{5\mathrm{MP}} = 2x_{5\mathrm{MP}} = 0.578 \text{ at/fiss} \qquad \text{[VB 3.1, 3.3]}$$

Now: sum of the five element systems "X in UO_2" (`SetSystem.C`), ENDF/B-VIII.0 yields with the fission fractions of an LWR cycle

$$\frac{dC_{\mathrm{FMP}}}{dt} = \sum_X y_X F, \quad X = \mathrm{Mo, Tc, Ru, Rh, Pd} \qquad \textbf{modified}, \text{ [VB 3.1]}$$

y_{Mo}	y_{Tc}	y_{Ru}	y_{Rh}	y_{Pd}	total
0.2370	0.0614	0.1442	0.0469	0.0771	0.5666 at/fiss

$$x_X = \frac{C_X^{\mathrm{prod}}}{\sum_X C_X^{\mathrm{prod}}} \quad \text{(42Mo-11Tc-25Ru-8Rh-14Pd at.\%)} \qquad \textbf{new}$$

$$\sum_X C_X^{\mathrm{prod}} = C_{m1} + C_{m2} + C^{\mathrm{intra}} + C^{\mathrm{GB}} \qquad \textbf{modified}, \text{ [VB 3.4]}$$

V. Bellani: $C_{5\mathrm{MP}} = C_m + C^{\mathrm{intra}} + C^{\mathrm{GB}}$. The particle composition is the production composition x_X (lumped species).

Grain-boundary zone

Zone 2 = shell of thickness δ inside each spherical grain of radius a, so that every boundary has a layer δ on both sides; zone 1 = grain interior.

$$\phi_2 = 1 - \left(1 - \frac{\delta}{a}\right)^3 \approx \frac{3\delta}{a} \ (\leq 0.999), \qquad \phi_1 = 1 - \phi_2, \qquad \delta = 100 \text{ nm} \quad \textbf{new}$$

$$P_k = \phi_k \, \Delta C_{\mathrm{prod}} \text{ (uniform production)}, \qquad c_k = \frac{C_{mk}}{\phi_k} \text{ (local concentration)}, \qquad k = 1, 2 \quad \textbf{new}$$

- δ = half-width of the grain-boundary regions analysed by TEM in the calibration (strips of about 200 nm across the boundary, thesis Figs. 5.3–5.16)
- calibration cases: $a = 12.3$ μm $\Rightarrow \phi_2 = 0.024$; ATM-109: $a = 15$ μm $\Rightarrow \phi_2 = 0.020$
- no new fitted parameter: δ comes from the TEM regions, the exchange from diffusion (below)

Equilibrium concentration and supersaturation

$$C_{\mathrm{eq}}(T) = C_{\mathrm{eq,ref}} \exp\left[-\frac{H_{\mathrm{sol}}^{\mathrm{eff}}}{k_B}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right], \qquad T_{\mathrm{ref}} = 2173.15\ \mathrm{K}$$ [VB 3.10, 3.11]

$$C_{\mathrm{eq,ref}} = x_{\mathrm{eq,ref}} \frac{4}{a_{\mathrm{UO_2}}^3} = 6 \times 10^{-5} \cdot 2.437 \times 10^{28} = 1.462 \times 10^{24}\ \mathrm{at/m^3}, \qquad H_{\mathrm{sol}}^{\mathrm{eff}} = 1.14\ \mathrm{eV}$$

[VB 3.12–3.15]

Mo solubility in UO_2 at 1900 °C (Kleykamp), $a_{\mathrm{UO_2}} = 547.5$ pm, DFT solution energy of Mo (Brillant et al.).

Supersaturation after the production of the time step:

$$S = \frac{C_m^n + \Delta C_{\mathrm{prod}}}{C_{\mathrm{eq}}(T)}$$ [VB 4.2, 4.3]

$$S_k = \frac{C_{mk}^n + \phi_k \Delta C_{\mathrm{prod}}}{\phi_k\, C_{\mathrm{eq}}(T)} = \frac{c_k^*}{C_{\mathrm{eq}}}, \qquad k = 1 \text{ for the intragranular, } k = 2 \text{ for the grain-boundary particles}$$

modified, [VB 3.16, 4.3]

$$\Delta\mu_k = k_B T \ln S_k \quad (S_k > 1;\ \text{no nucleation for } S_k \le 1)$$ [VB 3.17]

Nucleation: classical nucleation theory

Heterogeneous nucleation, capillarity approximation, j = intra, GB with S of the zone of the population:

$$\Delta G_j^* = \frac{B_j}{T^2(\ln S_k)^2}, \qquad B_j = \frac{16\pi\gamma_j^3\Omega_{5\mathrm{MP}}^2\Phi_j}{3k_B^2} \text{ (effective, calibrated)} \qquad \text{[VB 3.20, 3.21]}$$

$$k_{\mathrm{nucl},j} = k_{\mathrm{nucl},0,j}\exp\left(-\frac{\Delta G_j^*}{k_B T}\right), \qquad k_{\mathrm{nucl},0,j} = 1\ \mathrm{s}^{-1} \qquad \text{[VB 3.22]}$$

$$\frac{dN_j}{dt} = k_{\mathrm{nucl},j}\max\left(N_{\mathrm{site},j} - N_j, 0\right) \qquad \text{[VB 3.9]}$$

Backward Euler, N_j never decreases:

$$N_j^{n+1,*} = \frac{N_j^n + k_{\mathrm{nucl},j}N_{\mathrm{site},j}\Delta t}{1 + k_{\mathrm{nucl},j}\Delta t}, \qquad \Delta N_j = \max\left(N_j^{n+1,*} - N_j^n, 0\right) \qquad \text{[VB 4.5, 4.6]}$$

$B_{\mathrm{intra}} \neq B_{\mathrm{GB}}$, calibrated separately. V. Bellani: $B_{\mathrm{intra}} = B_{\mathrm{GB}} = B_{\mathrm{ref}}s_B$, $s_{B,\mathrm{intra}} = s_{B,\mathrm{GB}}$ [VB 6.2, 6.4].

Nucleation sites

Intragranular: dislocations (Veshchunov and Shestak) and intragranular fission gas bubbles

$$\rho_d = A\, b^n f_T, \qquad f_T = A_\infty + \frac{1 - A_\infty}{1 + \exp[(T - T_c)/\Delta T]}$$ [VB 3.23, 3.24]

$A = 6.545 \times 10^{12}$, $n = 1.151$, $A_\infty = 0.608$, $T_c = 1109$ K, $\Delta T = 25.8$ K, b in MWd/kgUO$_2$ [VB 3.25]

$$N_{\mathrm{site,intra}} = \phi_1 \left(f_d \frac{\rho_d}{\lambda} + f_b N_{b,\mathrm{intra}} \right), \qquad f_d = 0.67,\ f_b = 0.33,\ \lambda = 15 \text{ nm}$$ **modified**, [VB 3.29]

Factor ϕ_1: the intragranular sites are in zone 1 only. V. Bellani: no factor.

Grain boundary: the boundary surface, area per unit volume $3/(2a)$ (each boundary shared by two grains), one site every λ^2

$$N_{\mathrm{site,GB}} = \frac{A_{\mathrm{GB}}}{V} \frac{1}{\lambda^2} = \frac{3}{2a\lambda^2}$$ [VB 3.31–3.33]

End of r000 (54.1 MWd/kgUO$_2$): dislocation sites 2.2×10^{22}, bubble sites 1.4×10^{23}, $N_{\mathrm{intra}} = 5.8 \times 10^{21}$; GB sites 5.4×10^{20}, $N_{\mathrm{GB}} = 3.2 \times 10^{19}$ m^{-3} (4–6% of the sites occupied).

Metal taken by nucleation

Atoms per nucleus: critical size of CNT, at least a dimer (Ru clusters in UO_2)

$$n_{\mathrm{nucl},j} = \max\left(2, \frac{2\Delta G_j^*}{\Delta\mu}\right), \qquad \Delta C_{\mathrm{nucl},j} = n_{\mathrm{nucl},j}\,\Delta N_j \qquad \text{[VB 3.34–3.36, 4.7, 4.8]}$$

V. Bellani: the two requests are limited together by the dissolved metal, in proportion

$$\text{if } \Delta C_{\mathrm{nucl,intra}} + \Delta C_{\mathrm{nucl,GB}} > C_m^{\mathrm{avail}}: \quad f_{\mathrm{mass}} = \frac{C_m^{\mathrm{avail}}}{\Delta C_{\mathrm{nucl,intra}} + \Delta C_{\mathrm{nucl,GB}}}, \quad \Delta C_{\mathrm{nucl},j}^{\mathrm{act}} = f_{\mathrm{mass}}\Delta C_{\mathrm{nucl},j}$$

[VB 4.9–4.11]

Now: each population takes the atoms of its own zone ($k = 1$ for intra, $k = 2$ for GB)

$$\Delta C_{\mathrm{nucl},j}^{\mathrm{act}} = \min\left(n_{\mathrm{nucl},j}\Delta N_j,\ C_{mk}^n + \phi_k \Delta C_{\mathrm{prod}}\right) \qquad \textbf{modified}, \text{ [VB 4.9–4.11]}$$

$$\Delta N_j^{\mathrm{act}} = \frac{\Delta C_{\mathrm{nucl},j}^{\mathrm{act}}}{n_{\mathrm{nucl},j}}, \qquad N_j^{n+1} = N_j^n + \Delta N_j^{\mathrm{act}} \qquad \text{[VB 4.12]}$$

Growth (precipitation)

Diffusion-limited capture by spherical particles, $4\pi DR$ per particle (Ham; Brailsford et al.)

$$k_j = D_j\, 4\pi N_j R_j, \qquad D_j = k_{0,j} \exp\left(-\frac{\Delta G_j}{k_B T}\right), \qquad \Delta G_{\mathrm{intra}} = \Delta G_{\mathrm{GB}} = 1 \text{ eV} \qquad \text{[VB 3.38, 3.39]}$$

evaluated with the state at the beginning of the time step (no iteration on $R \to k \to C$) [VB 4.13].

$$J_{\mathrm{prec},j} = k_j \max\left(C_m - C_{\mathrm{eq}}, 0\right) \qquad \text{[VB 3.37]}$$

$$J_{\mathrm{prec,intra}} = k_{\mathrm{intra}}\left(c_1 - C_{\mathrm{eq}}\right), \qquad J_{\mathrm{prec,GB}} = k_{\mathrm{GB}}\left(c_2 - C_{\mathrm{eq}}\right) \qquad \textbf{modified}, \text{ [VB 3.37]}$$

each active only if its zone is supersaturated after production and nucleation, $c_k^* > C_{\mathrm{eq}}$ (V. Bellani: both active if $C_m^* > C_{\mathrm{eq}}$ [VB 4.19]). No thermal dissolution [VB Sec. 3.5].

$k_{0,\mathrm{intra}} \neq k_{0,\mathrm{GB}}$, calibrated separately: $D_{\mathrm{GB}}/D_{\mathrm{intra}} = 3.06$. V. Bellani: common k_0, $s_{k0,\mathrm{intra}} = s_{k0,\mathrm{GB}}$ [VB 6.5].

Exchange between the zones

Flux from zone 1 to zone 2 per unit fuel volume: $J_{\mathrm{ex}} = K_{\mathrm{ex}}(c_1 - c_2)$, two resistances in series: grain interior $\rightarrow$ interface (grain-boundary sink strength with the intragranular particles as internal sinks), interface $\rightarrow$ middle of zone 2 (distance $\delta/2$, area $3/a$ per unit volume).

$$K_{\mathrm{ex}} = \frac{D_{\mathrm{intra}}}{1/k_{\mathrm{GB}}^2 + a\delta/6}, \qquad k_{\mathrm{GB}}^2 = \max\left(k_1^2 \frac{F}{1-F}, \frac{15}{a^2}\right), \qquad F = \frac{3}{z}\left(\coth z - \frac{1}{z}\right), \quad z = k_1 a \quad \textbf{new}$$

$$k_1^2 = 4\pi \frac{N_{\mathrm{intra}}}{\phi_1} R_{\mathrm{intra}} \quad \text{(sink strength of the intragranular particles in zone 1)} \quad \textbf{new}$$

Equivalent Fick form: area $\times$ D / distance

$$K_{\mathrm{ex}} = \frac{3}{a}\frac{D_{\mathrm{intra}}}{L_1 + \delta/2}, \qquad L_1 = \frac{3}{a k_{\mathrm{GB}}^2} = \begin{cases} a/5 = 2.5\ \mu\mathrm{m} & \text{no particles (Booth sphere)} \\ \approx 1/k_1 & \text{particles dominate, } z \gg 1 \end{cases} \quad \textbf{new}$$

$1/k_1$ = distance travelled by a dissolved atom before the capture by a particle. r000: $L_1 = 107$ nm at 14 and 42 nm at 54.1 MWd/kgUO$_2$, against $\delta/2 = 50$ nm. No new parameter.

Exchange: derivation

1. Sphere of radius a, uniform source S, particles as uniform sinks k_1^2, interface concentration c_i, $u = c - c_i$, quasi-steady state:

$$D\,\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right) - Dk_1^2 u + S = 0, \quad u'(0) = 0,\ u(a) = 0$$

$$u(r) = \frac{S}{Dk_1^2}\left[1 - \frac{a\sinh(k_1 r)}{r\sinh(k_1 a)}\right]$$

2. Mean, with $\int_0^a r\sinh(k_1 r)\,dr = \frac{a\cosh z}{k_1} - \frac{\sinh z}{k_1^2}$, $z = k_1 a$:

$$\bar{u} = \frac{3}{a^3}\int_0^a u\,r^2\,dr = \frac{S}{Dk_1^2}(1 - F), \quad F = \frac{3}{z}\left(\coth z - \frac{1}{z}\right)$$

3. The source goes to the particles or across $r = a$: $S = Dk_1^2\bar{u} + Dk_{\mathrm{GB}}^2\bar{u}$

$$k_{\mathrm{GB}}^2 = \frac{k_1^2}{1-F} - k_1^2 = k_1^2\frac{F}{1-F}$$

4. Limits: $z \to 0$: $\coth z = 1/z + z/3 - z^3/45$, $F = 1 - z^2/15$, $k_{\mathrm{GB}}^2 \to 15/a^2$ (Booth sphere, $\bar{u} = Sa^2/15D$); $z \gg 1$: $F \approx 3/z$, $k_{\mathrm{GB}}^2 \to 3k_1/a$.

5. Interface $\to$ middle of zone 2, area $4\pi a^2/(\frac{4}{3}\pi a^3) = 3/a$:

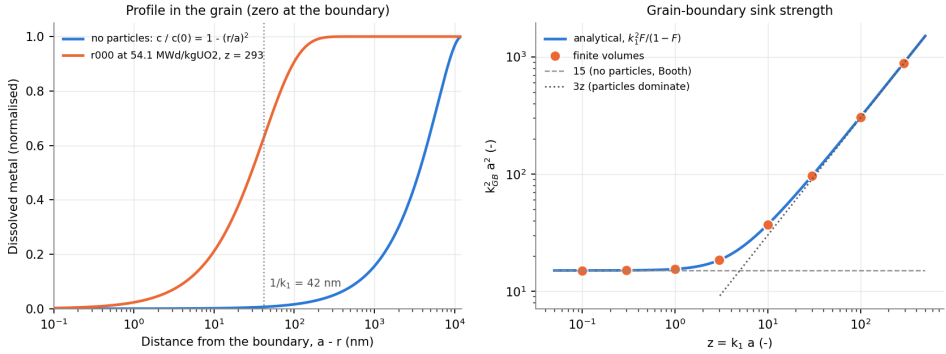
$$J = \frac{3}{a}\,\frac{D(c_i - c_2)}{\delta/2} = \frac{6D}{a\delta}(c_i - c_2)$$

6. Series: $c_1 - c_i = J/(Dk_{\mathrm{GB}}^2)$, $c_i - c_2 = Ja\delta/(6D)$, summing:

$$J = \frac{D(c_1 - c_2)}{1/k_{\mathrm{GB}}^2 + a\delta/6} = K_{\mathrm{ex}}(c_1 - c_2)$$

Exact: steps 1–4, for any c_i and with the C_{eq} term (only S changes).
Approximate: zone 2 well mixed with its mean at $\delta/2$; radius a instead of $a - \delta$ (0.8%); D_{intra} also in zone 2; quasi-steady state at each step. Code block marked [TBC]. Brailsford and Bullough (1981); Booth (1957).

Exchange: numerical check



- left: profile in the grain, without particles and with the particles of r000 at 54.1 MWd/kgUO$_2$ ($z = 293$): only the metal produced in the last $\sim$40 nm reaches the boundary
- right: $k_{\mathrm{GB}}^2 a^2$ from a finite-volume solution of step 1 vs the expression of step 3, agreement within 6×10^{-8} for $z = 0.1$–293, and the two limits of step 4 (`check_exchange.py`)

Irradiation-induced re-solution

First order in the precipitated inventory, same coefficient for the two populations

$$J_{\mathrm{res},j} = k_{\mathrm{res}}\, C^{j} \qquad \text{[VB 3.40]}$$

From the shrinkage of the FMP under ion irradiation (Sakib et al. 2025): $R_1/R_0 = 0.80/0.91$ between 0.30 and 1.75 dpa, $n \propto R^3$, $\dot{D}_{\mathrm{ref}} = 1$ dpa/day

$$k_{\mathrm{res}} = -\frac{3}{\Delta t} \ln \frac{R_1}{R_0}, \quad \Delta t = \frac{\Delta D}{\dot{D}_{\mathrm{ref}}} \;\Rightarrow\; k_{\mathrm{res,ref}} = 3.085 \times 10^{-6}\ \mathrm{s}^{-1} \qquad \text{[VB 3.41–3.45]}$$

$$k_{\mathrm{res}}(F) = k_{\mathrm{res,ref}} \frac{F}{F_{\mathrm{ref}}}, \qquad F_{\mathrm{ref}} = 1.48 \times 10^{19}\ \mathrm{fiss/(m^3\,s)} \qquad \text{[VB 3.46, 3.47]}$$

The re-solved metal returns to the zone of the particles: intragranular → zone 1, grain boundary → zone 2. V. Bellani: to the single matrix.

Metal balance: one matrix → two zones

V. Bellani, continuous form [VB 3.5–3.7] and backward-Euler 3×3 system, $\mathbf{C} = (C_m, C^{\mathrm{intra}}, C^{\mathrm{GB}})$ [VB 4.15–4.18]:

$$\frac{dC_m}{dt} = y_{5\mathrm{MP}} F - J_{\mathrm{nucl,intra}} - J_{\mathrm{nucl,GB}} - (k_{\mathrm{intra}} + k_{\mathrm{GB}})(C_m - C_{\mathrm{eq}}) + k_{\mathrm{res}}(C^{\mathrm{intra}} + C^{\mathrm{GB}}) \qquad \text{[VB 3.5]}$$

$$\begin{pmatrix} 1 + (k_{\mathrm{intra}} + k_{\mathrm{GB}})\Delta t & -k_{\mathrm{res}}\Delta t & -k_{\mathrm{res}}\Delta t \\ -k_{\mathrm{intra}}\Delta t & 1 + k_{\mathrm{res}}\Delta t & 0 \\ -k_{\mathrm{GB}}\Delta t & 0 & 1 + k_{\mathrm{res}}\Delta t \end{pmatrix} \mathbf{C}^{n+1} = \begin{pmatrix} C_m^* + (k_{\mathrm{intra}} + k_{\mathrm{GB}}) C_{\mathrm{eq}} \Delta t \\ C^{\mathrm{intra},*} - k_{\mathrm{intra}} C_{\mathrm{eq}} \Delta t \\ C^{\mathrm{GB},*} - k_{\mathrm{GB}} C_{\mathrm{eq}} \Delta t \end{pmatrix} \qquad \text{[VB 4.17, 4.18]}$$

Now, after production and nucleation, $\mathbf{C} = (C_{m1}, C_{m2}, C^{\mathrm{intra}}, C^{\mathrm{GB}})$:

$$\frac{dC_{m1}}{dt} = -k_{\mathrm{intra}}(c_1 - C_{\mathrm{eq}}) - K_{\mathrm{ex}}(c_1 - c_2) + k_{\mathrm{res}} C^{\mathrm{intra}} \qquad \textbf{modified}, \text{ [VB 3.5]}$$

$$\frac{dC_{m2}}{dt} = -k_{\mathrm{GB}}(c_2 - C_{\mathrm{eq}}) + K_{\mathrm{ex}}(c_1 - c_2) + k_{\mathrm{res}} C^{\mathrm{GB}} \qquad \textbf{new}$$

$$\frac{dC^{\mathrm{intra}}}{dt} = k_{\mathrm{intra}}(c_1 - C_{\mathrm{eq}}) - k_{\mathrm{res}} C^{\mathrm{intra}}, \qquad \frac{dC^{\mathrm{GB}}}{dt} = k_{\mathrm{GB}}(c_2 - C_{\mathrm{eq}}) - k_{\mathrm{res}} C^{\mathrm{GB}} \qquad \textbf{modified}, \text{ [VB 3.6, 3.7]}$$

production (P_k) and nucleation are applied before, as in [VB 4.2, 4.14].

Implicit 4×4 system

Backward Euler (`Solver::Laplace`), $\kappa_1 = k_{\mathrm{intra}}$ if $c_1^* > C_{\mathrm{eq}}$ else 0, $\kappa_2 = k_{\mathrm{GB}}$ if $c_2^* > C_{\mathrm{eq}}$ else 0 (* = after production and nucleation):

$$\begin{pmatrix} 1 + \frac{(\kappa_1 + K_{\mathrm{ex}})\Delta t}{\phi_1} & -\frac{K_{\mathrm{ex}}\Delta t}{\phi_2} & -k_{\mathrm{res}}\Delta t & 0 \\ -\frac{K_{\mathrm{ex}}\Delta t}{\phi_1} & 1 + \frac{(\kappa_2 + K_{\mathrm{ex}})\Delta t}{\phi_2} & 0 & -k_{\mathrm{res}}\Delta t \\ -\frac{\kappa_1\Delta t}{\phi_1} & 0 & 1 + k_{\mathrm{res}}\Delta t & 0 \\ 0 & -\frac{\kappa_2\Delta t}{\phi_2} & 0 & 1 + k_{\mathrm{res}}\Delta t \end{pmatrix} \begin{pmatrix} C_{m1} \\ C_{m2} \\ C^{\mathrm{intra}} \\ C^{\mathrm{GB}} \end{pmatrix}^{n+1} = \begin{pmatrix} C_{m1}^* + \kappa_1 C_{\mathrm{eq}}\Delta t \\ C_{m2}^* + \kappa_2 C_{\mathrm{eq}}\Delta t \\ C^{\mathrm{intra},*} - \kappa_1 C_{\mathrm{eq}}\Delta t \\ C^{\mathrm{GB},*} - \kappa_2 C_{\mathrm{eq}}\Delta t \end{pmatrix}$$

modified, [VB 4.17, 4.18]

- $1/\phi_k$: the fluxes depend on the local concentrations $c_k = C_{mk}/\phi_k$
- every column sums to 1: the total metal is conserved [VB 4.20]; checked on the outputs, inventory $= \sum_X C_X^{\mathrm{prod}}$ within 4×10^{-7} (the output precision)
- growth switched on per zone (V. Bellani: both on if $C_m^* > C_{\mathrm{eq}}$ [VB 4.19]); re-solution always active

Particle radius

Mean spherical-equivalent radius from the precipitated inventory and the number density

$$n_j = \frac{C^j}{N_j}, \qquad R_j = \left(\frac{3 n_j \Omega_{5\mathrm{MP}}}{4\pi}\right)^{1/3} \qquad \text{[VB 3.48, 3.49]}$$

$$\Omega_{5\mathrm{MP}} = \frac{V_w}{\eta_{\mathrm{hcp}}} = \frac{1.06651 \times 10^{-29}}{0.74} = 1.44123 \times 10^{-29}\ \mathrm{m^3/at} \qquad \text{[VB 3.50–3.52]}$$

(V_w: volumes of the atomic spheres, weights 40Mo:30Ru:10Tc:15Pd:5Rh, [VB Tab. 3.2]).

Now, only the metallic part of the inventory (with OpenCalphad), $R_j = 0$ while $N_j < 1\ \mathrm{m^{-3}}$:

$$R_j = \left(\frac{3\,\phi_{\mathrm{met}}\,C^j\,\Omega_{5\mathrm{MP}}}{4\pi N_j}\right)^{1/3}, \qquad \phi_{\mathrm{met}} = 1 - x_{\mathrm{Mo}}\,f_{\mathrm{ox}} \qquad \textbf{modified}, \text{ [VB 3.49]}$$

$\phi_{\mathrm{met}} = 1$ without thermochemistry; f_{ox} of the previous time step (next slide). The radius is used in k_j at the next step: $N_j, C^j \to R_j \to k_j \to C^j$ [VB Sec. 3.6].

Coupling with OpenCalphad at the grain boundary

With `iThermochimica`, OpenCalphad at the grain boundary receives the grain-boundary inventory of the model:

$$n_X^{\mathrm{OC}} = x_X\, C^{\mathrm{GB}}, \qquad X = \mathrm{Mo, Tc, Ru, Rh, Pd}$$ **new**

instead of the whole production "X produced", which was also counted in the FMP reservoirs (double inventory).

`JOGFormation` stores the fraction of this Mo in oxide phases (MoO_2, Mo_4O_{11}, ...):

$$f_{\mathrm{ox}} = \frac{\text{Mo in oxide phases}}{n_{\mathrm{Mo}}^{\mathrm{OC}}}, \qquad \phi_{\mathrm{met}} = 1 - x_{\mathrm{Mo}}\, f_{\mathrm{ox}} \text{ (all the particles)}$$ **new**

- the oxygen potential is uniform in the grain: f_{ox} of the grain boundary holds for all the particles
- the oxidised Mo stays in the particle inventory (re-solved with it); one time step of lag
- a step where OpenCalphad gives no result keeps the previous f_{ox}
- database `BaMoO_CsMoO_MoPdRhRuTc_merged.TDB` (Mo, O, Pd, Rh, Ru, Tc)

Oxygen balance between the fuel and the Mo oxides

UO_{2+x} with Kleykamp (1979) (`iStoichiometryDeviation = 7`, $\Delta x = 0.0013$ per % FIMA). The excess oxygen E is shared between the fuel and the Mo oxides of the particles (M = heavy-metal atoms per unit volume):

$$E = xM + O_{\mathrm{ox}}, \qquad \mathrm{Mo}_p = x_{\mathrm{Mo}}(C^{\mathrm{intra}} + C^{\mathrm{GB}}), \qquad \mathrm{Mo}_{\mathrm{GB}} = x_{\mathrm{Mo}} C^{\mathrm{GB}}$$ **new**

$E < 2\,\mathrm{Mo}_p$ — Mo/MoO_2 buffer: OpenCalphad with the fixed oxygen $E\,\mathrm{Mo}_{\mathrm{GB}}/\mathrm{Mo}_p$ gives μ_{O}, then x from Blackburn (Newton):

$$\ln \frac{p_{\mathrm{O_2}}}{1\ \mathrm{atm}} = 2\ln \frac{x(2+x)}{1-x} + 108x^2 - \frac{32700}{T} + 9.92 = \frac{2\mu_{\mathrm{O}}}{RT} + \ln \frac{1\ \mathrm{bar}}{1\ \mathrm{atm}}, \quad x \le \frac{E}{M}, \quad O_{\mathrm{ox}} = E - xM$$ **new**

$E \ge 2\,\mathrm{Mo}_p$ — all the Mo of the particles is MoO_2: $O_{\mathrm{ox}} = 2\,\mathrm{Mo}_p$, $x = (E - 2\,\mathrm{Mo}_p)/M$; OpenCalphad at the oxygen potential of the fuel.

Then x, $p_{\mathrm{O_2}}$, $\mu_{\mathrm{O_2}}$ of the fuel are updated (Blackburn) and ΔO_{ox} is removed from `O content`. Without the balance: x from Kleykamp and Mo oxidised without taking oxygen from the fuel. Only the Mo of the particles is an oxygen getter.

Sequence in a time step

1. production of the five systems (`FissionProductProduction`, `FissionProductDecay`) [per element]
2. ϕ_1, ϕ_2 from the grain radius; $P_k = \phi_k \Delta C_{\mathrm{prod}}$
3. $C_{\mathrm{eq}}(T)$; S_k, $\Delta\mu_k$ per zone
4. nucleation sites (ϕ_1 for the intragranular ones); ΔG_j^*, $k_{\mathrm{nucl},j}$ with B_j per population
5. ΔN_j (backward Euler), $n_{\mathrm{nucl},j}$, metal of nucleation limited by the zone
6. R_j, k_j from the state at the beginning of the step ($k_{0,j}$ per population); K_{ex}
7. 4×4 backward-Euler system: growth, exchange, re-solution
8. outputs: C's, N_j, R_j with ϕ_{met}, ϕ_2
9. with thermochemistry: `SetPhaseDiagram` (OpenCalphad at the grain boundary with $x_X C^{\mathrm{GB}}$ and the oxygen balance), `JOGFormation` $\rightarrow f_{\mathrm{ox}}$ for the next step

V. Bellani: steps 1–10 of thesis Sec. 4.3 and Fig. 4.1, routine `Simulation::MetallicFissionProducts()`, switch `iCm`. Now `FiveMetalParticles()`, switch `iFMP`.

Parameters

Parameter	Value	Origin
$k_{0,\mathrm{intra}}$	9.834342×10^{-16} m^2/s	calibrated (VB: 4.980×10^{-15}, common [VB 7.2])
$k_{0,\mathrm{GB}}$	3.009503×10^{-15} m^2/s	calibrated (VB: 4.980×10^{-15}, common)
B_{intra}	2.711740×10^{8} eV K^2	calibrated (VB: 2.666×10^{8}, common [VB 7.2])
B_{GB}	2.641409×10^{8} eV K^2	calibrated (VB: 2.666×10^{8}, common)
$\Delta G_{\mathrm{intra}} = \Delta G_{\mathrm{GB}}$	1.0 eV	V. Bellani [VB 6.9]
$k_{\mathrm{nucl},0,j}$	1.0 s^{-1}	V. Bellani [VB 6.9]
δ	100 nm	TEM grain-boundary regions (thesis Figs. 5.3–5.16)
$\Omega_{5\mathrm{MP}}$	1.44123×10^{-29} m^3/at	[VB 3.52]
n_{min}, λ	2 at, 15 nm	[VB 3.36, 3.27]
f_d, f_b	0.67, 0.33	image analysis [VB 3.30, 5.9]
$k_{\mathrm{res,ref}}$, F_{ref}	3.085×10^{-6} s^{-1}, 1.48×10^{19} fiss/(m^3 s)	Sakib et al. [VB 3.45, 3.47]
$x_{\mathrm{eq,ref}}$, $H_{\mathrm{sol}}^{\mathrm{eff}}$, T_{ref}	6×10^{-5}, 1.14 eV, 2173.15 K	Kleykamp, Brillant et al. [VB 3.11–3.15]
$A, n, A_\infty, T_c, \Delta T$	6.545×10^{12}, 1.151, 0.608, 1109 K, 25.8 K	Veshchunov and Shestak [VB 3.25]

Observables and objective

Projected area fraction in a TEM foil of thickness t_{TEM} (Poisson coverage)

$$f_{A,j} = 1 - \exp\left(-N_j \pi R_j^2 t_{\mathrm{TEM}}\right) \qquad \text{[VB 6.15, 6.16]}$$

$$f_{A,j} = 1 - \exp\left(-\frac{N_j}{\phi_k} \pi R_j^2 t_{\mathrm{TEM}}\right) \qquad \textbf{modified}, \text{[VB 6.16]}$$

with the number density in the zone of the population: the TEM grain-boundary images are strips around the boundary (zone 2), the intragranular images the grain interior. V. Bellani: N_{GB} per fuel volume gives $f_{A,\mathrm{GB}}$ 100–700 times below TEM, so it was left out [VB Sec. 6.3].

$t_{\mathrm{TEM}} = 50$, 50 (nominal), 70, 40 nm at $r/R = 0$, 0.30, 0.56, 0.75 [VB Sec. 6.3].

$$J_3 = E_{\mathrm{cal}} = \frac{1}{12} \sum_{i=1}^{12} \left| \ln \frac{M_i}{E_i} \right| \quad (f_{A,\mathrm{intra}}, R_{\mathrm{intra}}, R_{\mathrm{GB}}) \qquad \text{[VB 6.17, 6.18]}$$

$$J_4 = \frac{1}{16} \sum_{i=1}^{16} \left| \ln \frac{M_i}{E_i} \right| \quad (f_{A,\mathrm{intra}}, R_{\mathrm{intra}}, R_{\mathrm{GB}}, f_{A,\mathrm{GB}}) \qquad \textbf{new}$$

Model values at the end of the history (54.1 MWd/kgUO$_2$); TEM data of [VB Tab. 5.2].

Calibration

V. Bellani [VB Ch. 6, 7]

- screening of 9 scaling factors, then 2 free: s_k (log), s_B (linear), common to the two populations
- 512 Sobol points + 1 reference point, best sampled point (no optimiser) [VB 6.10–6.19]
- $s_k = 4.9803 \times 10^{-15}$, $s_B = 266.5765$, $E_{\mathrm{cal}} = 0.614$ [VB 7.2, 7.3]

Two-zone model (`calibration_two_zones/`)

- 4 free: $k_{0,\mathrm{intra}}$ ($\times 10^{\pm 2}$, log), B_{intra} ($\times$0.5–1.5), $k_{0,\mathrm{GB}}$ ($\times 10^{-1}$–10^{3}, log), B_{GB} ($\times$0.5–1.5)
- 4096 scrambled Sobol points (seed 42) + 5 Nelder-Mead refinements from the best points: all end at $J_4 = 0.5541$–0.5544
- copy of the code with the parameters from the environment, no OpenCalphad
- B's and $k_{0,\mathrm{GB}}$ well identified; $k_{0,\mathrm{intra}}$ less ($\times$0.155–0.197 for the same J_4)

	V. Bellani	two zones
$k_{0,\mathrm{intra}}$ (m^2/s)	4.98×10^{-15}	9.83×10^{-16}
B_{intra} (eV K^2)	2.666×10^{8}	2.712×10^{8}
$k_{0,\mathrm{GB}}$ (m^2/s)	4.98×10^{-15}	3.010×10^{-15}
B_{GB} (eV K^2)	2.666×10^{8}	2.641×10^{8}
J_3 (this model)	0.616	0.598
J_4	0.869	**0.554**

Check, 4 Nov 2026: the calibration copy is identical to `src/`, `include/` except the 4 parameter lines; $J_4 = 0.5541$ recomputed; $\pm 5\%$ on each parameter always increases J_4:

	$\times$0.95	$\times$1.05
$k_{0,\mathrm{intra}}$	0.5596	0.5594
B_{intra}	0.6120	0.6356
$k_{0,\mathrm{GB}}$	0.5629	0.5588
B_{GB}	0.6641	0.6782

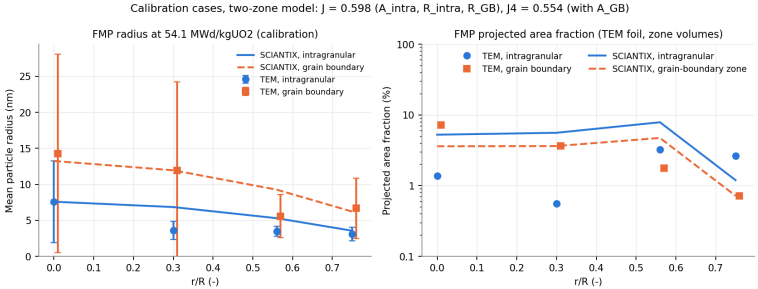
Calibration cases: UO_2, PWR, 54.1 MWd/kgUO_2

Gravelines 5 PWR standard UO_2, five cycles, section burnup 61.4 GWd/tU, TEM data of Angleraud (2024) at $r/R = 0$, 0.30, 0.56, 0.75 [VB Sec. 5.1, Tab. 5.1, 5.2]. Histories reconstructed by V. Bellani [VB Sec. 5.4, Tab. 5.6]:

Cycle	Duration (h)	F (fiss/m^{3}s)	T_0 (K)	$T_{0.30}$ (K)	$T_{0.56}$ (K)	$T_{0.75}$ (K)
1	6000	1.48×10^{19}	1171	1123	1003	869
2	6000	1.60×10^{19}	1210	1158	1029	886
3	6000	1.52×10^{19}	1181	1132	1009	873
4	6000	1.31×10^{19}	1111	1068	961	842
5	6000	1.32×10^{19}	1113	1070	963	843

- same fission rate at the four positions (scaled by $\alpha = 1.132$ to 54.1 MWd/kgUO_2 [VB 5.19–5.22]); temperature from the ALCYONE profile of the last cycle scaled with the linear power [VB 5.10–5.14]
- settings of the cases: grain radius 12.3 μm, 4.0% U-235, no He, no HBS, `iStoichiometryDeviation = 0`, `iThermochimica = 0` (inputs of V. Bellani, `regression/test_Cm`)
- regression: `python3 -m testing.runner --metallicinclusions`

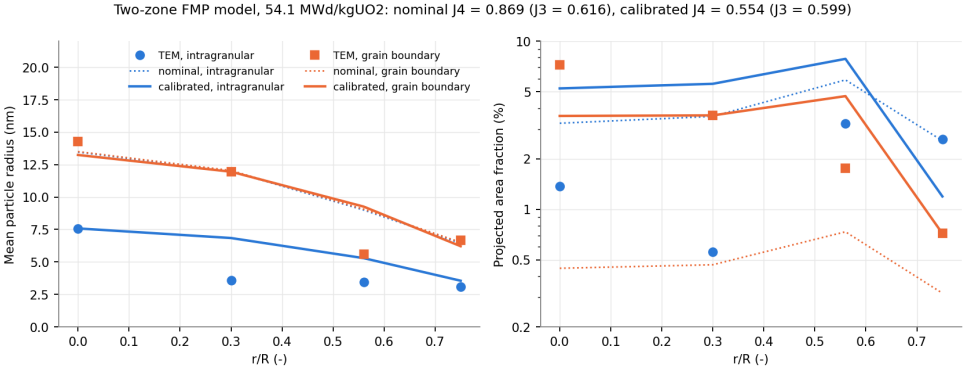
Calibration cases: results



r/R	R_{intra} (nm) TEM / model	R_{GB} (nm) TEM / model	$f_{A,\mathrm{intra}}$ (%) TEM / model	$f_{A,\mathrm{GB}}$ (%) TEM / model	metal dissolved	metal in GB particles
0.00	7.58 / 7.58	14.28 / 13.24	1.37 / 5.24	7.21 / 3.59	14.2%	2.45%
0.30	3.59 / 6.84	11.95 / 11.95	0.56 / 5.58	3.62 / 3.62	17.6%	2.23%
0.56	3.45 / 5.29	5.58 / 9.26	3.25 / 7.86	1.76 / 4.72	35.2%	1.62%
0.75	3.10 / 3.55	6.68 / 6.21	2.62 / 1.19	0.72 / 0.72	88.8%	0.28%

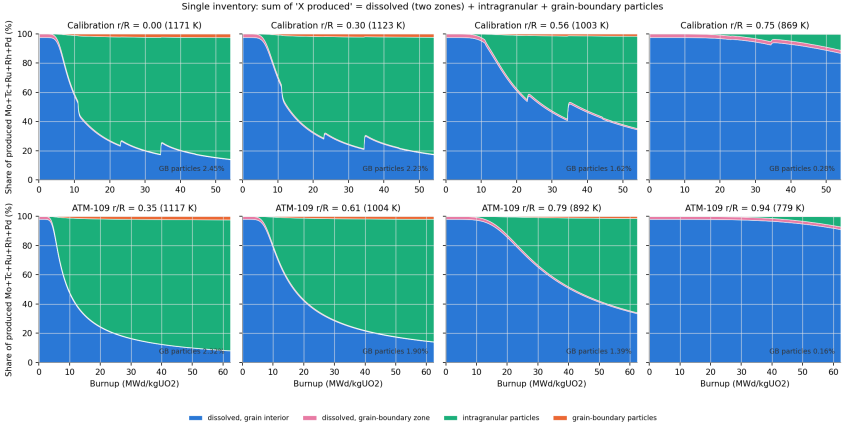
$J_4 = 0.554$, $J_3 = 0.598$. With the parameters of V. Bellani: $R_{\mathrm{intra}} \approx R_{\mathrm{GB}} = 13.5$, 12.0, 9.0, 6.4 nm, $f_{A,\mathrm{GB}} = 0.3$–0.7%, $J_4 = 0.869$.

Calibration: parameters of V. Bellani vs calibrated



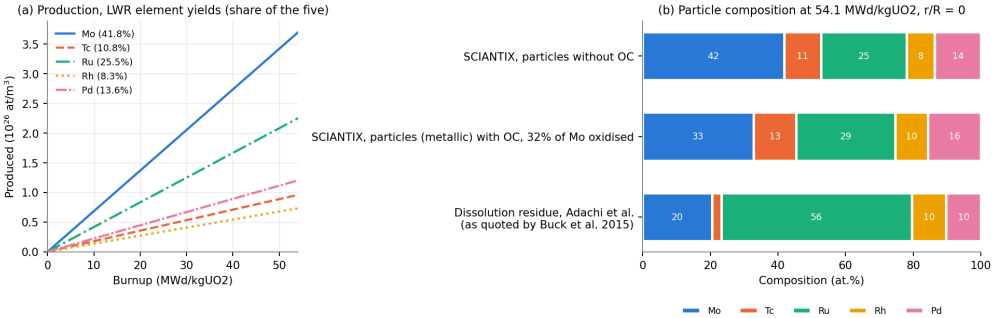
- $R_{\mathrm{GB}}/R_{\mathrm{intra}}$ from 1.00 to 1.75: the faster grain-boundary growth separates the radii; R_{intra} now 1.0–1.9 times TEM (was 1.8–3.3)
- $f_{A,\mathrm{GB}}$ within a factor 2 at three positions (was 2–16 times too low); too high at $r/R = 0.56$
- $f_{A,\mathrm{intra}}$ worse at $r/R = 0$–0.30 (4–10 times TEM): smaller but more numerous intragranular particles

Where the metal goes



Top: calibration cases; bottom: ATM-109 rings. Most of the metal precipitates in the intragranular particles above ∼1000 K; at $r/R = 0.75$ (869 K) 89% stays dissolved. The grain-boundary particles hold 0.2–2.5%.

Elements: production and particle composition



(a) Production of the five systems, LWR yields: 41.8% Mo, 10.8% Tc, 25.5% Ru, 8.3% Rh, 13.6% Pd.
(b) Particle composition = production composition; with OpenCalphad 32% of the Mo is oxidised (r000) and the metallic particles are 33Mo-13Tc-29Ru-10Rh-16Pd. Dissolution residue quoted by Buck et al. (2015, p. 237): 20Mo-0.5/5Tc-50/60Ru-10Rh-10Pd.

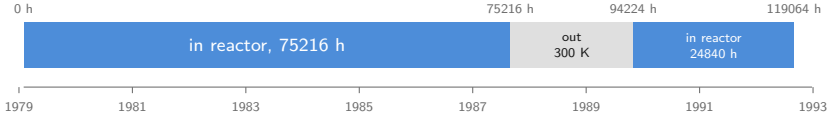
Independent validation: Buck et al. (2015), ATM-109

E.C. Buck, E.J. Mausolf, B.K. McNamara, C.Z. Soderquist, J.M. Schwantes, *Nanostructure of metallic particles in light water reactor used nuclear fuel*, J. Nucl. Mater. 461 (2015) 236–243. Not used in the calibration.

Information	Where in the paper
ATM-109, Quad Cities I BWR, GE UO_2, 3.0% U-235, initial grain size 30 μm	Sec. 1, p. 237 (Wolf et al. [21])
in reactor Feb 1979 – Sep 1987 and Nov 1989 – Sep 1992; specimens maximum burnup 70.80 GWd/MTU	Sec. 1, p. 237
cooling ∼21 years (July 2013); residues of nitric acid and carbonate-peroxide dissolutions	Sec. 2.1, p. 237
LWR centreline below 1200 °C, rim 400–500 °C	Sec. 1, p. 237
residue composition 20Mo:0.5–5Tc:50–60Ru:10Rh:10Pd (Adachi et al.)	Sec. 1, p. 237
SEM/TEM of the residues, hcp ε-Ru, crystallites of 1–3 nm	Secs. 2.2, 3, pp. 238–240
round particles of 10–300 nm in diameter	abstract, p. 236; Sec. 5, p. 242
cumulative *number* distribution of the particle volume, log-normal fits	Fig. 6, Sec. 3, p. 240
mean particle size 43 nm (carbonate-peroxide), 58 nm (nitric)	Sec. 3, p. 240

No power or temperature history in the paper (it would need the ATM-109 reports, Guenther et al. [10], Wolf et al. [21]) and no radial position (particles from dissolved fuel) ⇒ assumed history, next slide.

ATM-109: irradiation history (assumed)



- continuous operation in the two periods of the paper (p. 237), 100056 h in reactor; between them $F = 0$, 300 K
- constant $F = 5.003 \times 10^{18}$ fiss/(m^3 s) $\Rightarrow$ 62.41 MWd/kgUO$_2$ = 70.8 GWd/tHM at the end: 70.80 GWd/MTU is the *maximum* burnup of the specimens, so an upper bound
- four rings of equal volume, $(r/R)^2 = 0.125$, 0.375, 0.625, 0.875, uniform F (no rim peak)
- parabolic profile $T(r) = T_s + (T_c - T_s)[1 - (r/R)^2]$, $T_c = 900\,^\circ$C (below the 1200 °C of p. 237), $T_s = 450\,^\circ$C (rim 400–500 °C, p. 237), constant in time
- grain radius 15 μm (grain size 30 μm), 3.0% U-235; other settings as the calibration cases

Case	r/R	T (K)	T (°C)
Buck2015_r035	0.354	1117	844
Buck2015_r061	0.612	1004	731
Buck2015_r079	0.791	892	619
Buck2015_r094	0.935	779	506

cases `test_FMP_UO2_Buck2015_*`
regression group `--metallicinclusions`

ATM-109: how the model is compared with Fig. 6

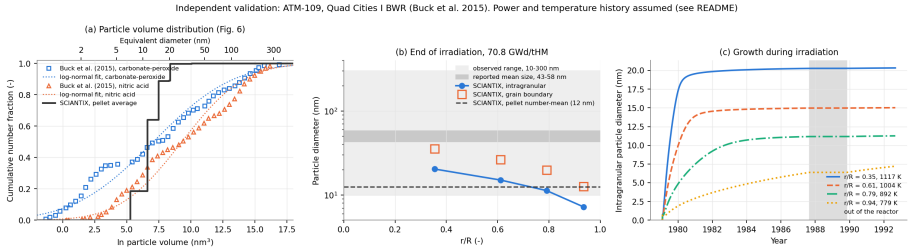
- Fig. 6 (p. 240): cumulative number fraction vs $\ln V$ (V in nm^3) for the two dissolutions; digitized (`data/buck2015/fig6_*.txt`, binned marker pixels, approximate)
- log-normal fit of $\ln V$ (`data/buck2015/summary.txt`); diameter of the equivalent sphere $d = (6V/\pi)^{1/3}$:

$$\ln d \sim \mathcal{N}(\mu_d, \sigma_d), \quad \mu_d = \frac{\mu_{\ln V} + \ln(6/\pi)}{3}, \quad \sigma_d = \frac{\sigma_{\ln V}}{3}, \quad d_{\mathrm{median}} = e^{\mu_d}, \quad \bar{d} = e^{\mu_d + \sigma_d^2/2}$$

- the fits give number-mean diameters of 50 and 62 nm, close to the 43 and 58 nm of the text: the reported values look like number means
- model: pellet-averaged population, the four rings with equal volume, intragranular and grain-boundary particles weighted by number (N_{intra}, N_{GB}); one size per ring and population

	$\mu_{\ln V}$, $\sigma_{\ln V}$	median d (nm)	number-mean d (nm)
carbonate-peroxide	7.07, 4.90	13	50 (reported 43)
nitric acid	9.30, 3.83	28	62 (reported 58)

ATM-109: results



Ring	d_{intra} (nm)	d_{GB} (nm)	metal dissolved	metal in GB part.
0.35, 1117 K	20.3	35.5	8%	2.32%
0.61, 1004 K	15.0	26.2	14%	1.90%
0.79, 892 K	11.3	19.7	34%	1.39%
0.94, 779 K	7.2	12.6	93%	0.16%

	median d (nm)	mean d (nm)	mean ln V (V in nm^3)
Buck, carbonate-peroxide	13	50 (43)	7.07
Buck, nitric acid	28	62 (58)	9.30
SCIANTIX, two zones	11	12.5	6.78
SCIANTIX, VB parameters	19	18.5	7.95

Median just below the carbonate-peroxide series; number mean 3.5–5 times too low; distribution much narrower (one size per ring and population), the 100–300 nm particles are missing; GB particles (26–36 nm in the inner rings) closer, but 2% of the particles.

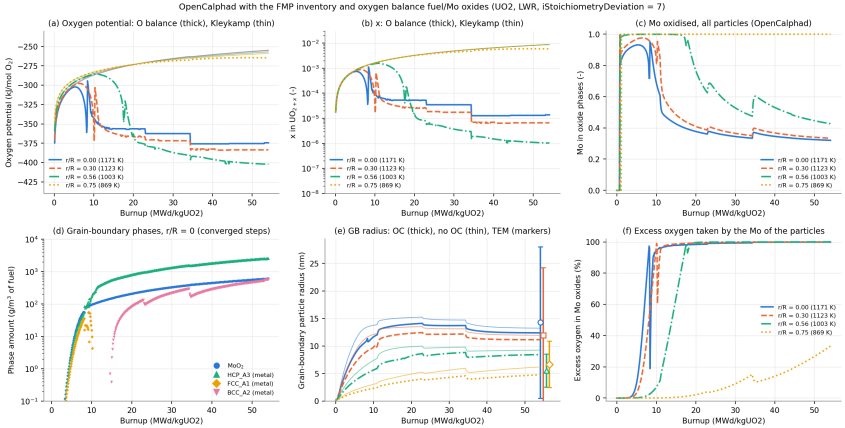
Thermochemistry cases: setup and results

`thermochemistry/test_FMP_UO2_OC_*`: the calibration cases with `iThermochimica = 2` (OpenCalphad at every step), `iStoichiometryDeviation = 7` (Kleykamp 1979, same x(burnup) at the four positions), Blackburn oxygen potential, $p = 10^5$ Pa. Regression group `--metallicinclusions-oc --oc`.

r/R	μ_{O_2} (kJ/mol)	x	Mo oxidised	R_{intra} (nm) OC / no OC	R_{GB} (nm) OC / no OC	R_{GB} (nm) TEM
0.00	−375	1.4×10^{-5}	32%	7.1 / 7.6	12.4 / 13.2	14.3
0.30	−384	6.6×10^{-6}	33%	6.4 / 6.8	11.2 / 12.0	12.0
0.56	−402	1.0×10^{-6}	43%	4.8 / 5.3	8.5 / 9.3	5.6
0.75	−265	5.8×10^{-3}	100%	2.7 / 3.6	4.8 / 6.2	6.7

- $r/R = 0$–0.56: fuel buffered at Mo/MoO_2, x far below Kleykamp (8.7×10^{-3}, −255 kJ/mol, all Mo oxidised without the balance)
- $r/R = 0.75$: 89% of the metal dissolved, the particle Mo is not enough, all oxidised; x rises with burnup
- OpenCalphad: MoO_2, HCP_A3 and BCC_A2 (metal), Mo_4O_{11} at some steps; at a few steps no convergence (previous f_{ox} kept); in r000 the convergence of one step can change between runs (final values within 10^{-5})

Thermochemistry cases: evolution



(a, b) oxygen potential and x with the oxygen balance (thick) and Kleykamp only (thin); (c) Mo oxidised; (d) phases at the grain boundary; (e) R_{GB} with and without OpenCalphad vs TEM; (f) share of the excess oxygen in the Mo oxides.

Verification, 4 November 2026

- **Regression** (`python3 -m testing.runner --metallicinclusions --metallicinclusions-oc --oc`): **12 passed, 0 failed** — 4 calibration cases, 4 ATM-109 rings, 4 thermochemistry cases; build up to date with the sources
- **Calibration still valid**: model of the calibration copy = current code (only the 4 parameter lines differ); $J_4 = 0.5541$ recomputed with the parameters in the code; $\pm 5\%$ on each parameter increases J_4 (local minimum)
- **Mass conservation**: $C_{m1} + C_{m2} + C^{\mathrm{intra}} + C^{\mathrm{GB}} = \sum_X C_X^{\mathrm{prod}}$ within 4×10^{-7} (output precision) in r000, r075, ATM-109 r035
- **Exchange**: k_{GB}^2 analytical vs finite volumes within 6×10^{-8} (`check_exchange.py`)
- all the changes tagged `EC November 2026` in the code; documentation in `validation/metallicinclusions/README.md`

Known limits and open points

- $f_{A,\mathrm{intra}}$ 4–10 times TEM at $r/R = 0$–0.30, 2 times too low at 0.75: the intragranular number density is the main remaining discrepancy
- R_{GB} too large at $r/R = 0.56$ (9.3 vs 5.6 nm)
- at low temperature most of the metal stays dissolved (89% at $r/R = 0.75$, 93% in the outer ATM-109 ring)
- one mean size per population: no size distribution; ATM-109 mean size underestimated (12.5 vs 43–58 nm)
- K_{ex} [TBC]: zone 2 as a well-mixed layer, D_{intra} also in zone 2
- **inputs inherited from V. Bellani**: grain radius 12.3 μm and 4.0% U-235, while thesis Tab. 5.1 reports a mean grain size of 8.8 μm and 4.48% U-235 (a enters ϕ_2, $N_{\mathrm{site,GB}}$, K_{ex}: a change needs a new calibration)
- calibration without OpenCalphad (the Mo oxidation reduces the radii by 5–25%)
- oxygen balance: only the particle Mo takes oxygen, excess oxygen from Kleykamp, no radial O/M
- lumped species: particle composition = production composition (measured particles poorer in Mo, richer in Ru)
- MOX systems keep fast-reactor yields: validated only for UO_2 in LWR conditions

References

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